Outlook

P-RAG and Factual RAG - my notes/research

From Vasudeva Varma <vv@iiit.ac.in>
Date Thu 9/4/2025 10:55 AM
To Prasoon Dev <prasoon.dev@research.iiit.ac.in>

Study the following in detail and see the aspects of Factual retrieval part in the context of learning.

Thanks,
-Vasu

There are three systems actually: P-RAG, L-RAG and Factual-Retrieval:

Factual Retrieval: Factual knowledge resides in Long Term Memory (LTM); Working Memory (WM) retrieves it when needed.

Longitudinal Retrieval: WM asks for "activated LTM" from learner history across sessions.

Pedagogical Retrieval: Pedagogical strategies (e.g., spacing) are stored in LTM and activated for task-specific sequencing

Factual Retrieval (*mostly RAG*)

addressing core LLM limitations like hallucination and static internal knowledge

- **Building Subject Matter Experts (SMEs): Data Creation and Keeping SMEs up-to-date**

- **Pre-retrieval: Query processing, expansion, ...**
- **Post-retrieval: Reranking, summarization, and fusion optimizations**

- **Improving efficiency in knowledge intensive tasks like Open domain QA/ dialogue**

Factual Retrieval

Building Subject Matter Experts

- **Parametric vs non-parametric memory? Hybrid?**
  - **Adaptive retrieval based on query characteristics[Mallen et al ACL 2023]**
  - **Empirically, more reliance on RAG for context[F&J EMNLP 2024]**
- **Trainable key-value memory layers**
  - **Memory Layers at Scale[Berges et al Meta FAIR 2024]**
  - **Fact-Injected Language Model (FILM)[Verga et al NAACL 2021]**

- **Episodic Memory Controllers (Larimar)[Das et al IBM 2024]**
- **RAG with Heterogeneous Memory (MATTER)[Lee et al Amazon AGI 2024]**
- **Improving efficiency in knowledge intensive tasks like Open domain QA/dialogue: EMAT [Wu et al EMNLP 2023]**

Memory-augmented Large Language Models (LLMs) are a new architectural paradigm designed to overcome the inherent limitations of models with **fixed-length context windows**. By integrating an **external memory module**, these models gain the ability to store, update, and retrieve information from past interactions, allowing them to "remember" long-term context far beyond their standard token limit. This approach decouples the model's main processing from its long-term storage, using **explicit memory banks** that can be continuously updated. This evolution enhances performance and opens new frontiers in natural language understanding and generation, leading to more adaptable and context-aware AI systems.

The major memory augmentation techniques and the problems they solve include:

- **External Memory Modules / Explicit Memory Banks (General Concept)**
  - **What they are**: These systems integrate an external memory component with the core LLM, functioning like a **dynamic notepad** where key information is cached and recalled to enrich responses. This involves **decoupling the model's main processing from its long-term storage**, moving from implicit memory (embedded in parameters) to explicit memory banks. Techniques like memory compression and efficient retrieval algorithms are employed to keep the external memory manageable.
  - **Problems solved**:
    - Overcoming the **inherent limitations of fixed-length context windows**.
    - Enabling models to handle tasks such as **multi-turn conversations, document summarization, and complex reasoning** that require sustained context over time.
    - Allowing the model to **learn and recall information over extended periods**, similar to human working memory.
- **Retrieval-Augmented Generation (RAG)**
  - **What it is**: RAG is a paradigm where an LLM is **coupled with a retrieval system to augment its inputs with relevant external information**. Instead of relying solely on static "memorized" knowledge, a RAG system **retrieves documents or facts on the fly** based on the query and feeds them into the model's context before generation. This typically involves **external memory indexing** (e.g., a **vector index** of textual knowledge, databases, file systems, or APIs) and a retrieval mechanism. The external memory can

be continuously updated, making it "dynamic".
    - **Problems solved**:
        - Accessing **up-to-date or domain-specific information** that was not in the LLM's original training data.
        - Mitigating **hallucinations** and addressing the issue of **temporal knowledge cutoff** by grounding responses in retrieved, verifiable evidence.
        - Handling **proprietary or internal data** securely by connecting LLMs to an organization's specific knowledge bases.
        - Ensuring responses are **grounded in truth** and explicitly refer to sources, which increases trust and allows for verification.
        - Addressing **stale knowledge** by allowing updates to the external memory without requiring retraining of the LLM itself.
        - Serving as a **cost optimization** by reducing the need for ultra-large models or extensive fine-tuning, as a smaller model with RAG can perform like a much bigger one on domain-specific tasks.
- **Dynamic Knowledge Graphs (KGs)**
    - **What they are**: KGs are **structured networks of entities and relationships** that encode enterprise knowledge. When integrated with LLMs (often as part of a **Graph-augmented RAG or GraphRAG**), the LLM's retrieval step uses **graph queries or walks** to fetch related information, beyond just free-text embeddings. KGs can be updated in real-time, making them a **dynamic, queryable memory**.
    - **Problems solved**:
        - Enabling **more precise reasoning on structured facts**.
        - Providing **precision** by grounding the LLM with exact facts, helping to virtually eliminate hallucination, which is critical in regulated industries.
        - Enhancing **recall** by allowing the LLM to handle unstructured text while the KG covers structured information.
        - Supporting reasoning over the **most current structured knowledge**, especially valuable in domains with frequently changing data like finance or engineering.
        - Helping to **synthesize complex queries** that involve multiple linked pieces of data, which traditional RAG might struggle to do from separate passages.
- **Architectural Memory Modules**
    - **What they are**: These involve integrating memory directly into the transformer architecture itself, often using **latent caches or learnable key-value stores**.
    - **Problems solved** and examples:
        - **MemReasoner (2025)**: Introduces a transformer with a **separate latent memory module** to iteratively read/update context. It demonstrated **better generalization on synthetic reasoning tasks** (e.g., bAbI logical QA) compared to standard transformers by explicitly storing facts and learning their temporality.
        - **Memory Fusion / MemLong (2024)**: Avoids feeding overly large retrieved text by injecting it as **key-value vectors at upper transformer layers** instead of raw text. This effectively **extends usable context** (e.g., from 4k tokens to ~80k tokens) and **preserves generation quality** even with enormous contexts.

- **MemoryLLM (2024) and M+ (2025)**: *MemoryLLM* pioneered **latent-space memory** by compressing past context into a 1B-parameter memory pool. Its extension, *M+*, combines a **co-trained retriever with latent memory** to dynamically fetch relevant facts. *M+* achieves dramatic gains in **retaining knowledge up to 160k tokens of context** with similar GPU usage, outperforming prior models that struggled beyond 20k tokens.
- **Memory Layers at Scale (Dec 2024)**: Introduced **trainable key-value memory layers** that add billions of external parameters **without increasing inference cost**. A model augmented with these layers outperformed a dense transformer with more than twice the compute on knowledge-intensive tasks, achieving **higher accuracy on QA benchmarks** than much larger standard models by offloading factual recall to a large external memory.
- **MATTER (Memory-Augmented Transformer for heterogeneous knowledge) (2024)**: Retrieves from **multiple knowledge sources** (unstructured text and QA pairs) into a **fixed-size neural memory**. This enables **much faster inference** (~100x higher throughput) while matching conventional retrieve-and-read QA accuracy.

These memory-augmented approaches fundamentally enhance LLMs' capabilities by addressing their limitations regarding context length, factual accuracy, knowledge retention, and real-time information access, making them significantly more viable for real-world enterprise applications.

Factual retrieval ensures that AI-powered educational applications provide accurate and up-to-date information, addressing core LLM limitations like hallucination and static internal knowledge. Retrieval-Augmented Generation (RAG) is a primary technique here.

- RAG Workflow and Advantages: RAG combines retrieval systems with LLMs, integrating vector-based information retrieval with generative models to dynamically include external, task-specific knowledge.
  - How it Works: The LLM interprets the user's question, the query is matched against a document repository (information retrieval), and the retrieved data is fed to the LLM for contextual and grounded generation.
  - Key Advantages: RAG enhances accuracy by incorporating real-world, dynamic data, reduces hallucinations by limiting reliance on outdated or inferred knowledge, and provides domain-specific adaptability, crucial for specialized educational content (e.g., healthcare, legal).
- Components and Design Principles: Key components of RAG include a Retriever (e.g., BM25, Dense Passage Retrieval using BERT embeddings) and a Generator. Knowledge sources can be structured (databases) or unstructured (documents). Design principles emphasize selecting appropriate retrieval methods, ensuring high-quality and up-to-date knowledge sources, and optimizing for efficiency without compromising accuracy.
- Building Subject Matter Experts (SMEs): RAG is essential for building effective SMEs, which are AI agents possessing holistic and detailed knowledge of a specific subject (e.g., high school biology, operating

systems). This involves:

◦

Data Creation for SME through RAG: Synthetic data generation using information retrieval techniques like query expansion and query translation based on Named Entity Recognition (NER) and Multi-Word Expressions (MWE).

◦

Keeping SME Up-to-Date: RAG inherently supports dynamic knowledge updates, addressing the challenge of maintaining current information within SMEs.

•

Advanced RAG Architectures: Beyond "Naive RAG," "Advanced RAG" involves pre-retrieval (query routing, rewriting, expansion) and post-retrieval (reranking, summarization, fusion) optimizations. "Modular RAG" combines multiple modules like search, predict, rewrite, rerank, and fusion.

•

Efficiency in Knowledge-Intensive Tasks: Models like EMAT (Efficient Memory-Augmented Transformer) demonstrate how external knowledge can be encoded into a key-value memory and queried efficiently using Maximum Inner Product Search (MIPS) for knowledge-intensive NLP tasks such as open-domain question answering and dialogue. EMAT's architecture allows memory retrieval to run concurrently with the transformer's forward pass, significantly reducing computational overhead. This approach can outperform traditional parametric models in accuracy and speed, even with fewer resources.

•

Challenges: Key issues for factual retrieval include ensuring retrieval relevance and data quality, addressing ethical concerns (bias and privacy), and achieving scalability in real-time applications. Hallucination, while mitigated by RAG, can still arise from flawed data sources, inferior data utilization, or issues during pre-training and inference.

improving **factual accuracy and knowledge capacity** of LLMs via learnable memory. Meta AI's layers at large

Mallen et al ACL 2023] https://aclanthology.org/2023.acl-long.546.pdf
[F&J EMNLP 2024] Farahani, Johansson https://aclanthology.org/2024.emnlp-main.943/
[V et al, NAACL, 2021] https://aclanthology.org/2021.naacl-main.288.pdf
Wu et al EMNLP 2023 https://aclanthology.org/2022.emnlp-main.346.pdf
Das et al IBM 2024 https://arxiv.org/pdf/2403.11901
Lee at al, 2024 https://arxiv.org/pdf/2406.04670

Memory augmentation techniques have proven highly effective for factual learning, especially when building subject matter experts (SMEs) or domain-specific intelligent systems. These techniques enable models to access, update, and reason over large volumes of factual knowledge dynamically—features essential for accurate, up-to-date, and domain-specific learning.
Key Memory Augmentation Techniques

1. Trainable Key-Value Memory Layers

- Description: External memory modules can be added to language models as key-value stores, where keys are learned embeddings and values represent factual information. These layers are sparsely activated, allowing efficient lookup and update.
- Effectiveness: Meta AI's *Memory Layers at Scale* demonstrated that models with 128B memory slots outperformed much larger dense transformers on knowledge-intensive QA benchmarks, despite using less compute.
- Datasets:
  - NaturalQuestions: Open-domain factoid QA benchmark.
  - WebQuestionsSP: Knowledge-intensive QA dataset.
- Methods:
  - Memory Layers at Scale: Trainable key-value memory layers integrated into LLMs.
  - Fact Injected Language Model (FILM): Incorporates a neuro-symbolic knowledge base (KB) as a fact memory, improving QA performance by up to 27 points on WebQuestionsSP over open-book baselines.

2. Episodic Memory Controllers

- Description: Episodic memory modules, such as Larimar, allow models to store and update contextual or user-specific information in real time without retraining.
- Effectiveness: Larimar enables fast, accurate, and gradient-free updates to memory, reducing hallucinations and supporting selective forgetting or censoring of sensitive information.
- Datasets:
  - Custom user interaction logs (for personalized learning).
  - QA benchmarks with evolving facts.
- Methods:
  - Larimar: Episodic memory controller for LLMs, supporting one-shot updates and robust contextual learning.

3. Retrieval-Augmented Generation (RAG) and Heterogeneous Memory

- Description: Models like MATTER retrieve relevant facts from multiple knowledge sources (text, QA pairs) and encode them into a compact, queryable memory representation.
- Effectiveness: MATTER matches conventional RAG accuracy but achieves ~100× higher throughput, making it suitable for scalable SME applications.
- Datasets:
  - PAQ (Probably Asked Questions): Large-scale QA dataset.
  - NaturalQuestions, TriviaQA: Standard QA benchmarks.
- Methods:
  - MATTER: Memory-augmented transformer using heterogeneous knowledge sources.
  - Efficient Memory-Augmented Transformer (EMAT): Outperforms RePAQ on PAQ, demonstrating strong generative capabilities for factual QA.

4. Non-Parametric Memory and Adaptive Retrieval

- Description: Non-parametric memories (e.g., vector databases) store factual information externally and retrieve it as needed. Adaptive retrieval strategies decide when to retrieve based on the query's characteristics.
- Effectiveness: Retrieval-augmented LMs outperform much larger LMs on long-tail factual QA, especially for less popular entities. Adaptive retrieval further improves efficiency by only retrieving when necessary.
- Datasets:
    - POPQA: Entity-centric QA dataset (14k questions).
    - WebQuestionsSP, NaturalQuestions.
- Methods:
    - Adaptive Retrieval: Retrieves from non-parametric memory based on entity popularity and relationship types, improving both accuracy and efficiency.
    - Fact Injected Language Model (FILM): As above, but with explicit integration of KB facts.

5. Memory-Augmented LLMs for Reasoning

- Description: Architectures like MemReasoner learn the relative order of facts in context, supporting multi-hop reasoning and temporal dependence.
- Effectiveness: MemReasoner, with supporting fact supervision, achieves near-perfect accuracy on multi-hop reasoning tasks like BABILong, where standard LLMs struggle.
- Datasets:
    - bAbi, BABILong: Synthetic reasoning benchmarks.
- Methods:
    - MemReasoner: Memory-augmented LLM with explicit supervision on supporting facts.

Pedagogical Retrieval

better learning processes and enhance meta-learning

- **Bloom's 2-sigma problem through personalisation**

- **Hybrid approach for Structured tutoring: <<Pump -> Hint -> Prompt -> Assertion>> strategy by MWPTutor**

- **Empowering LLMs as Meta-Cognitive partners**
  - **Chunking**
  - **Conquering procrastination**
  - **Memory enhancement**
  - **Personalised Learning and strategy discovery**
  - **Assessment, Progress Monitoring and reflection**

Pedagogical retrieval involves integrating insights from learning sciences to design AI systems that not only deliver information but also facilitate effective learning processes, enhancing meta-learning and providing structured tutoring.

•

Addressing the Bloom's 2 Sigma Problem: One-on-one tutoring is highly effective but difficult to scale. LLMs offer a path to scale personalized tutoring by providing personalized instruction, immediate feedback, active engagement, and assessment for mastery.

•

Hybrid Approach for Structured Tutoring: Pure LLM tutors may leak answers or follow poor teaching strategies. A hybrid approach, like MWPTutor (Math Word Problems Tutor), combines human-designed pedagogy with LLM content generation. This preserves pedagogical structure (e.g., breaking problems into logical steps, using a "Pump → Hint → Prompt → Assertion" strategy) while leveraging LLMs to write tutor scripts and generate content dynamically. MWPTutor Live, which is contextual and adaptive, outperforms a cached model and freeform GPT-4 in tutoring quality, although hints can still be redundant.

•

LLMs as Metacognitive Partners: LLMs, when thoughtfully integrated, have the potential to significantly enhance metacognitive skills—"learning how to learn"—across all age groups. This supports various meta-learning principles:

◦

Chunking: LLMs can break down complex topics into digestible segments, identify core concepts, and simplify explanations, aiding in managing cognitive load.

◦

Renaissance Learning: LLMs' vast interdisciplinary knowledge allows them to explore connections between fields, provide resources for diverse subjects, and facilitate cross-disciplinary problem-solving,

supporting the development of versatile understanding.

◦

Conquering Procrastination: LLMs can help break down large tasks, create study schedules, set realistic goals, and suggest coping strategies.

◦

Memory Enhancement: LLMs can generate flashcards, assist in scheduling spaced repetition, explain concepts in different ways, and help create memory aids for better knowledge retention.

◦

Personalized Learning & Strategy Discovery: LLMs can analyze individual learning styles and knowledge gaps to create personalized learning pathways, adapt content difficulty, and recommend resources. They can also introduce and help learners reflect on various learning techniques (e.g., active recall, Feynman technique).

◦

Progress Monitoring & Reflection: LLMs can provide feedback, identify areas for improvement, ask reflective questions, help visualize progress, and assist in creating concept maps.

•

AI Assistants for Content Creation and Assessment: LLMs can generate lesson plans, activities, questions, feedback, and even assess essays. For example, in Second Language (L2) assessment, GPT-4 has shown promise in conducting analytic assessments of writing, correlating significantly with linguistic features. While holistic scoring is easier, LLMs are efficient for more challenging tasks like analytic assessment.

Vasudeva Varma | Professor |

IIIT Hyderabad | Gachibowli, Hyderabad 500032, India |

vv@iiit.ac.in | +91 40 6653 1179 | www.iiit.ac.in/~vv |

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY
HYDERABAD

IIIT Hyderabad